

Weekly Internship Report — User Management System (Week 1)

INTERNSHIP REPORT Week 1 — User Management System

18 May 2026 – 24 May 2026

Intern	Nisarg Jignesh Shah
Project Title	User Management System
Tech Stack	ASP.NET Core MVC PostgreSQL Entity Framework Core Bootstrap
Report Type	Weekly Internship Report
Week Covered	Week 1

Week 1, Day 1 — 18 May 2026

Project Setup, MVC Architecture & PostgreSQL Integration

1. Intern Information

Field	Details
Intern Name	Nisarg Jignesh Shah
Date	18 May 2026
Week / Day	Week 1 / Day 1
Project Title	User Management System
Tech Stack	ASP.NET Core MVC
Role	Software Developer Intern
Reporting To	Hemal Shah (Manager)
Work Mode	On-Site
Company	Ncode Solutions (GNFC)
University	Pandit Deendayal Energy University

Field	Details
Company Name	Ncode Solutions, GNFC
Company Address	(n)Code Solutions (A Division of GNFC Ltd.), 14th Floor, Tower One, Road 5C, Zone 5, Gujarat International Finance Tec-City, Gandhinagar, Gujarat 382050
Internship Role	Software Developer Intern
Internship Mode	On-Site
Reporting Manager	Hemal Shah (Manager)
Internship Duration	18 May 2026 to 12 Jul 2026
Project Title	User Management System
Report Type	Weekly Internship Report
Week Covered	Week 1

2. Day Objective

The objective of Week 1 Day 1 was to understand the ASP.NET Core MVC architecture, configure the development environment, establish PostgreSQL database connectivity, design the initial database schema for role-based authentication, and begin development of the User Management System.

3. Work Completed

During the first week of my internship at Ncode Solutions, GNFC, I worked on the development of a User Management System using ASP.NET Core MVC framework. The objective of the project is to create a secure and responsive web application that supports user authentication, role-based authorization, user management functionalities, and dashboard operations.

This week mainly focused on understanding the project architecture, setting up the development environment, configuring the database connection, designing the database schema, and implementing the core authentication functionalities.

3.1 ASP.NET Core MVC Project Initialization

- Created a new ASP.NET Core MVC web application project structure

- Understood Controllers, Models, Views, ViewModels, and Data layer organization
- Configured Program.cs and middleware pipeline
- Integrated Bootstrap 5 for responsive frontend design
- Set up Razor View rendering structure and layout configuration

3.2 PostgreSQL Database Integration

- Installed PostgreSQL and configured local database server
- Connected ASP.NET Core MVC application with PostgreSQL using Npgsql Entity Framework provider
- Configured connection string inside appsettings.json
- Created ApplicationDbContext for database operations
- Verified successful database connectivity and runtime operations

3.3 Database Schema Design

- Created RoleMaster table with Admin and User roles
- Created UserMaster table with role-based foreign key relationship
- Added fields such as FullName, Username, Password, EmailId, MobileNo, DOB, Gender, and CreatedDate
- Implemented relational mapping using Entity Framework Core
- Performed insert, select, and validation queries for testing

3.4 Authentication Module Development

- Developed user registration functionality with form validation
- Created login system using Username and Password verification
- Implemented session-based authentication using HttpContext.Session
- Added role-based dashboard navigation logic
- Restricted UserList page access to Admin users only

3.5 Responsive UI Development

- Designed responsive Register and Login pages using Bootstrap cards and forms
- Created dynamic navigation bar with role-based menu rendering
- Implemented responsive dashboard cards for statistics display
- Added conditional rendering for Login/Register and Logout/UserList menus
- Improved overall user interface consistency and layout responsiveness

3.6 Admin Dashboard & User List

- Developed Admin Dashboard with user statistics cards

- Added Total Users, Total Admins, and Normal Users count cards
- Created responsive User List table using Bootstrap table components
- Implemented role badges and action buttons for Edit/Delete operations
- Added search functionality for Full Name, Username, Mobile Number, and Role

3.7 Toast Notifications & User Experience Improvements

- Implemented Bootstrap Toast notification system
- Added success notifications for Registration, Login, Logout, and Delete operations
- Used TempData for cross-request notification handling
- Improved frontend interaction and professional dashboard feel
- Enhanced navbar and dashboard styling with gradients and hover effects

4. Task Status Summary

#	Task	Category	Status
1	ASP.NET Core MVC project setup	Backend	COMPLETED
2	PostgreSQL database configuration	Database	COMPLETED
3	Entity Framework Core integration	Backend	COMPLETED
4	RoleMaster table creation	Database	COMPLETED
5	UserMaster table creation	Database	COMPLETED
6	Foreign key relationship implementation	Database	COMPLETED
7	Registration module	Backend	COMPLETED

8	Login authentication system	Backend	COMPLETED
9	Session-based authentication	Security	COMPLETED
10	Role-based dashboard access	Backend	COMPLETED
11	Responsive Login/Register UI	Frontend	COMPLETED
12	Dynamic navbar implementation	Frontend	COMPLETED
13	Admin dashboard cards	Frontend	COMPLETED
14	User list management page	Frontend	COMPLETED
15	User search functionality	Backend	COMPLETED
16	Toast notification system	Frontend	COMPLETED
17	Bootstrap responsive styling	UI/UX	COMPLETED
18	Database validation and testing	Testing	COMPLETED

5. Key Learnings

#	Topic
---	-------

1	ASP.NET Core MVC
2	C#
3	Entity Framework Core
4	PostgreSQL
5	Bootstrap 5
6	Razor Views
7	Session Management
8	HTML5
9	CSS3
10	JavaScript
11	Visual Studio Code

- Learned practical implementation of ASP.NET Core MVC architecture and request lifecycle
- Understood Entity Framework Core integration with PostgreSQL databases
- Learned session-based authentication and role-based authorization concepts
- Gained experience with Bootstrap responsive UI development
- Learned how Razor Views dynamically render data from controllers
- Understood LINQ queries for filtering and searching user data
- Learned how TempData is used for toast notifications across redirects
- Improved debugging skills for configuration, namespace, and database-related errors
- Learned responsive admin dashboard design and Bootstrap component usage

Confidential — Internship Report | Week 1, Day 1

Week 1, Day 2 — 19 May 2026

Authentication Improvements, Dashboard Enhancements & Search Features

1. Intern Information

Field	Details
Intern Name	Nisarg Jignesh Shah

Date	19 May 2026
Week / Day	Week 1 / Day 2
Project Title	User Management System
Tech Stack	ASP.NET Core MVC
Role	Software Developer Intern
Reporting To	Hemal Shah
Work Mode	On-Site

2. Day Objective

The objective of Day 2 was to improve authentication flow, enhance the dashboard user interface, implement dynamic search functionality, and strengthen the admin user management experience.

3. Work Completed Today

3.1 Navbar & Layout Redesign

- Developed dynamic navbar with role-based rendering
- Guest users can see Login and Register buttons only
- Logged-in Admin users can access Dashboard, User List, and Logout
- Logged-in normal users can access Dashboard and Logout
- Added Bootstrap gradient navbar styling and hover effects

3.2 Search Functionality

- Added dynamic user search system using LINQ queries
- Implemented search by Full Name, Username, Mobile Number, and Role
- Added dropdown filter selection for better usability
- Used AsQueryable() for flexible query construction
- Optimized admin user filtering workflow

3.3 Dashboard UI Improvements

- Added professional statistics cards with responsive layout
- Displayed Total Users, Admins, and Normal Users counts dynamically
- Added role-based colored badges inside User List table
- Improved Bootstrap table responsiveness for smaller devices

- Added hover effects and card shadows for modern UI appearance

3.4 Toast Notification Integration

- Integrated Bootstrap Toast notification component globally in Layout page
- Configured TempData notification rendering for redirects
- Added success notifications for Login, Logout, Registration, and Delete operations
- Implemented JavaScript auto-show toast behavior

3.5 Bug Fixes & Debugging

- Resolved ViewModel namespace resolution issues
- Fixed duplicate compile item problems in project configuration
- Corrected Razor layout rendering issues
- Fixed appsettings.json formatting errors
- Resolved PostgreSQL database naming and connection issues

4. Task Status Summary

#	Task	Category	Status
1	Dynamic navbar implementation	Frontend	COMPLETED
2	Role-based navigation rendering	Security	COMPLETED
3	LINQ-based search functionality	Backend	COMPLETED
4	Search filter dropdown	Frontend	COMPLETED
5	Responsive dashboard cards	UI/UX	COMPLETED
6	User count analytics cards	Frontend	COMPLETED
7	Toast notification system	Frontend	COMPLETED

8	TempData notification integration	Backend	COMPLETED
9	Razor layout fixes	Frontend	COMPLETED
10	Namespace and ViewModel fixes	Backend	COMPLETED
11	PostgreSQL connection troubleshooting	Database	COMPLETED
12	Bootstrap UI enhancement	Frontend	COMPLETED

5. Key Learnings

- Learned advanced Razor conditional rendering techniques
- Understood dynamic query generation using LINQ and IQueryable
- Learned TempData lifecycle and notification rendering flow
- Improved debugging and troubleshooting skills in ASP.NET Core MVC
- Learned responsive admin dashboard layout structuring using Bootstrap Grid
- Understood session-based role authorization handling in layouts

6. Conclusion

During the first week of internship at Ncode Solutions (GNFC), substantial progress was made in developing the User Management System using ASP.NET Core MVC and PostgreSQL.

The work completed included authentication system development, database integration, responsive UI design, dashboard analytics, search functionality, and admin user management features. The internship provided strong practical exposure to full-stack web application development and significantly improved backend, frontend, and debugging skills.

Submitted By

Nisarg Jignesh Shah

Software Developer Intern

Pandit Deendayal Energy University

Weekly Summary — Week 1

The major objectives completed during the first week are:

1. Understanding ASP.NET Core MVC project structure.
2. Setting up PostgreSQL database connectivity.
3. Learning Entity Framework Core integration.
4. Designing database tables for role and user management.
5. Implementing user registration and login functionality.
6. Implementing session-based authentication.
7. Creating role-based dashboard functionality.
8. Designing responsive user interface using Bootstrap.
9. Implementing search functionality in user listing.
10. Adding professional UI enhancements such as dashboard cards and toast notifications.

Work Performed During Week 1

1. Project Setup and Environment Configuration

The project environment was configured using ASP.NET Core MVC framework. Required packages related to Entity Framework Core and PostgreSQL provider were installed successfully.

The application structure including Controllers, Models, Views, ViewModels, and Database Context was organized properly according to MVC architecture.

2. PostgreSQL Database Integration

A PostgreSQL database was configured and connected with the ASP.NET Core MVC application.

The following database operations were performed:

- Database creation
- Connection string configuration
- Entity Framework Core setup
- DbContext configuration
- Table mapping
- Data insertion and retrieval

The database schema consisted of the following tables:

RoleMaster

Stores different user roles such as:

- Admin
- User

UserMaster

Stores complete user details including:

- Full Name
- Username
- Password
- Email ID
- Mobile Number
- Date of Birth
- Gender
- Role Information
- Created Date

Foreign key relationships between the tables were implemented successfully.

3. User Registration Module

A complete registration module was developed where users can:

- Register new accounts
- Select roles
- Enter personal details
- Store user data in database

Validation was implemented using ViewModels and model validation attributes.

The registration form was designed using Bootstrap for responsive behavior.

4. User Login System

A secure login module was developed with:

- Username and password validation
- Database verification
- Session creation after login

- Authentication logic
- Invalid credential handling

Role-based redirection functionality was also implemented.

5. Session Management

Session-based authentication was implemented to maintain user login states.

The following session functionalities were developed:

- Username session storage
- Role session storage
- Session validation
- Logout functionality
- Role-based access control

6. Admin Dashboard Development

A professional admin dashboard UI was designed with:

- Dashboard cards
- User statistics
- Role-wise user count
- Responsive layout
- Bootstrap styling

The dashboard provides quick access to:

- User List
- Dashboard Home
- Logout

7. User List Management

An admin-only user listing page was implemented where administrators can:

- View all registered users
- View role details
- Search users
- Manage user records

The table was designed using Bootstrap responsive table components.

8. Search Functionality

Advanced search functionality was implemented with filtering options based on:

- Full Name
- Username
- Mobile Number
- Role

Search operations were implemented using LINQ queries and Entity Framework Core.

9. UI and Frontend Improvements

Several UI enhancements were implemented including:

- Responsive navbar
- Role-based navigation menu
- Dashboard cards
- Bootstrap styling
- Professional layouts
- Responsive design
- Sidebar dashboard layout

The application was optimized for better user experience.

10. Toast Notification System

Toast notification functionality was implemented to provide:

- Login success messages
- Registration success messages
- Logout notifications
- Delete confirmation messages

Bootstrap toast components and TempData were used for notification handling.

Challenges Faced

During the development process, several technical issues were encountered and resolved.

Challenges:

- PostgreSQL database connection errors
- MVC routing and view configuration issues

- Session management setup
- ViewModel namespace resolution issues
- JSON configuration errors in appsettings.json
- Entity Framework package conflicts
- Navbar and layout rendering problems
- Database migration and table mapping issues

Solutions Implemented:

- Corrected connection strings and database names
- Fixed namespace and folder structure problems
- Configured session middleware properly
- Updated layout and Razor view configurations
- Corrected JSON formatting issues
- Resolved duplicate compile item conflicts
- Implemented proper Bootstrap integration

Learning Outcomes

During the first week of internship, I gained practical experience in:

- ASP.NET Core MVC architecture
- Database connectivity with PostgreSQL
- Entity Framework Core integration
- Session-based authentication
- MVC Controllers and Views
- Razor syntax
- Bootstrap responsive design
- Role-based authorization
- LINQ query implementation
- UI/UX improvements
- Debugging and error handling

The internship helped improve both backend and frontend development skills.

Future Scope

The following functionalities are planned for future implementation:

- Password hashing and encryption
- Edit and Delete user functionality
- Profile management
- Chart-based analytics dashboard
- Dark mode UI
- AJAX-based live search
- Pagination
- Email notifications
- Profile image upload
- Activity logs
- Export to PDF and Excel

Conclusion

The first week of internship at Ncode Solutions, GNFC provided valuable exposure to real-world web application development using ASP.NET Core MVC.

The User Management System project helped in understanding practical implementation of authentication systems, database integration, responsive UI design, and role-based access control.

The tasks completed during this week significantly improved my understanding of full-stack web development concepts and professional software development practices.

Submitted By

Nisarg Jignesh Shah

Software Developer Intern

Pandit Deendayal Energy University